



Strategic Intelligence Gateway for Next-Gen Analytics & Lending

<https://signalwebsite.onrender.com/>

<https://signal-aaq.onrender.com/>

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AGENDA



What We'll Cover

- 1 The Problem**
Why manual credit monitoring doesn't scale
- 2 What SIGNAL Is (and Isn't)**
Scope, purpose, and firm boundaries
- 3 How It Works**
Pipeline, rule logic, and the severity model
- 4 Trust & Governance**
Data model, security, and known risks
- 5 Status & Roadmap**
Where the project stands, and why it matters



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The Problem

- Every quarter, Credit Analysts read interim financial statements, calculate ratios, and compare them against covenant thresholds — for every borrower in the portfolio.
- That process does not scale. As portfolios grow, so does the manual workload.
- Manual, judgment-based review is inconsistent: two analysts under time pressure can review the same file and reach different conclusions.
- Warning signs caught late leave the institution and the borrower with fewer options — and a manual process is hard to prove was done consistently for governance and regulators.

IN ONE SENTENCE

Credit teams need a scalable, consistent, explainable way to catch early warning signs of borrower financial distress — without losing the human judgment that credit decisions ultimately require.



How SIGNAL Adds Value

Question / Metric	Value Proposition / Impact
What's the big picture?	SIGNAL is envisioned as a fully AI-assisted commercial credit monitoring platform
Why is it worth doing?	Operational efficiency, governance, earlier risk detection, increased analyst capacity
How much money does it save?	Estimated \$400K–\$550K per annum in productivity improvements
How much time does it save?	Estimated 60–70% time savings: From 7–8 hours manual, down to 2.5 - 3 hours
How many more loans/reviews?	Estimated 50–60% increase in loan review volume
What does increased throughput mean?	More borrower reviews, more proactive engagement, better portfolio monitoring, less loan losses.
What happens next?	Enterprise integration, continuous monitoring, explainable AI.





02 · SCOPE



What SIGNAL Is

SIGNAL is a proof-of-concept, AI-assisted decision-support system for commercial credit monitoring. It ingests borrower financial documents in whatever format they arrive, calculates the same ratios a credit analyst would compute by hand, checks them against a bank's own Early Warning Indicator rules, and raises a flagged, explainable alert when something looks off.

Reads any format

PDFs, spreadsheets, images, scanned statements, emails

Calculates like an analyst

The same ratios, against the bank's own EWI rules

Explains every alert

Source document, extracted value, and rule version attached

Every decision stays with a human.





What SIGNAL Deliberately Does Not Do

This boundary is the most important design decision in the whole project — it is intentional, not a limitation to be removed later.

- ✗ It does not approve or decline credit.
- ✗ It does not assign official credit ratings or risk scores.
- ✗ It does not predict borrower default.
- ✗ It does not recommend a borrower course of action, restructuring, or remediation. It advises the Credit Analyst on available next steps within the lending institution's ecosystem.

SIGNAL's job stops at surfacing relevant, well-evidenced information efficiently. Interpretation, risk judgment, and action remain entirely with qualified human professionals.

Synthesizing Something that Looks Close to Reality

Generating a Synthetic Data Set is effectively the opposite of what SIGNAL proposes to do. Two public datasets are combined into a clean master borrower record, then used to generate semi-realistic customer documents through RAG and an LLM, with controlled variation by role, type, and format. The LLM handles the formatting, the RAG tells it what to include and/or omit by type and role, and the information contained in the master data remains intact. There are no made-up numbers, roles, or companies.



The Process ensures realistic, varied, and purpose-built documents while preserving the integrity of the underlying master data. Every generated document is traceable back to its source record and generation parameters.

How SIGNAL Works

SIGNAL follows the same basic path a human analyst follows, automated end to end, with every step visible and explainable:



- Ingest**
Accept borrower packages in mixed formats — PDF, Excel, CSV, images, text/email.
- Extract**
Pull out and store key metrics such as revenue, EBITDA, debt, working capital, covenant terms, key dates.
- Validate**
Check extracted data for completeness, consistency, plausibility.
- Analyze**
Calculate ratios/submit to spreading and test against EWI rules and covenant thresholds.
- Alert**
Surface anything that crosses a threshold, rated Low / Medium / High.
- Present**
Show it in a role-appropriate dashboard view for review and action.

Every step is logged. Every alert traces back to a source document, an extracted value, and a named rule version — so the output can always be explained and defended, not just trusted.

Under the Hood: Eight Stages

Underneath those six steps, the pipeline is actually built as eight discrete, auditable stages — each with a defined input and output.



The rule logic in stages 6 and 7 is intentionally kept deterministic rather than left to a model to infer — so any result can be reproduced and defended later.

The Obvious Question: Why a hybrid architecture?

SIGNAL uses NLP/CV/ML where interpretation is required, deterministic logic where calculations and policy rules must be reproducible, and human review where evidence is ambiguous or professional judgment is required. More complex Deep Learning, LLM, and Foundation Models were not warranted against the complexity of the problem presented.

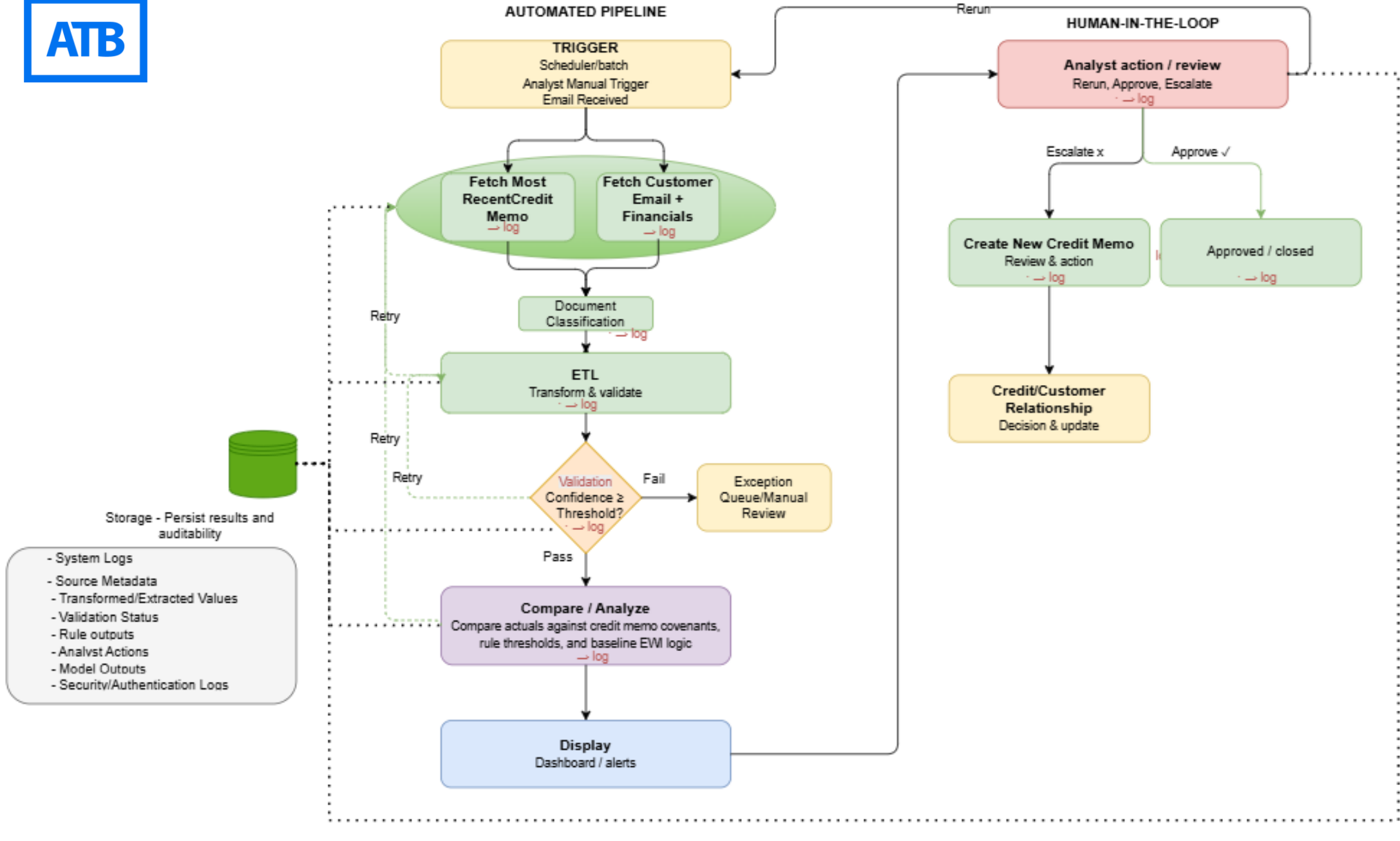
What SIGNAL Analyzes

FINANCIAL RATIO CATEGORIES

- Profitability — revenue growth, gross margin, EBITDA margin, net margin
- Liquidity — current ratio, quick ratio, working capital
- Leverage — debt/tangible net worth, debt/EBITDA, debt/assets
- Coverage — interest coverage, DSCR (current & pro forma)
- Efficiency — A/R days, inventory days, A/P days, cash conversion cycle
- Cash flow — operating cash flow / debt

EARLY WARNING INDICATOR RULES

- Late financial reporting
- Leverage covenant breach
- DSCR / current ratio covenant breach
- Financial statement age
- Credit bureau-style proxy score
- Adverse document language
- Risk rating context (trend only)



Data Model & Traceability

Nine relational tables carry the system end to end, built on a single SQLite file. Every result traces back to where it came from:



EVERY RECORD CARRIES

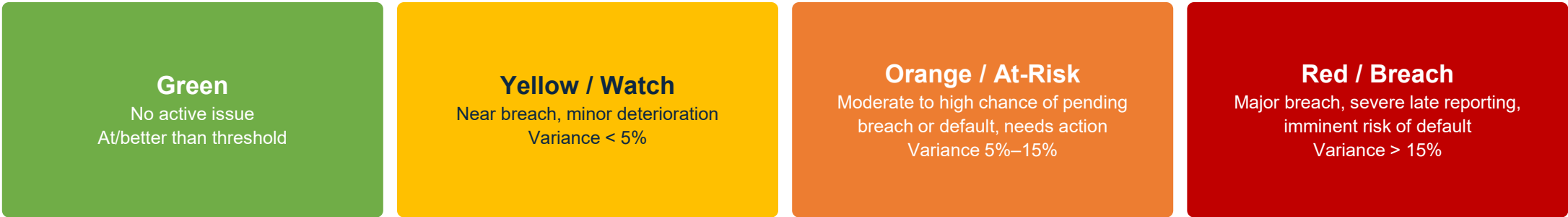
job_id	source_document_id	rule_version_id	model_version_id	confidence score
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So any dashboard figure can be traced back to the exact document, rule version, and extraction run that produced it.



The Severity Model

Every trigger — a covenant breach or a document-language flag — is scored on the same four-level scale:



Every triggered item maps to a recommended next action for the analyst — a workflow suggestion only. SIGNAL does not take the action itself and does not make the underlying credit decision.





04 · TRUST & GOVERNANCE

Who It's For: Role-Based Access

SIGNAL is a role-based application: what a person can see and do is scoped to their role, and access narrows as you move down.

This reflects the current live build's 'sign-in' role picker — three roles, different dashboard presentations, one shared underlying system.

Administrator

Full operational & governance access — scheduling, system logs, rule configuration, system operations monitoring, user management.

Credit Manager

Document access, expanded analyst view, management workspace, decision audit logs, rule changes, and the broader business workflow.

Credit Analyst

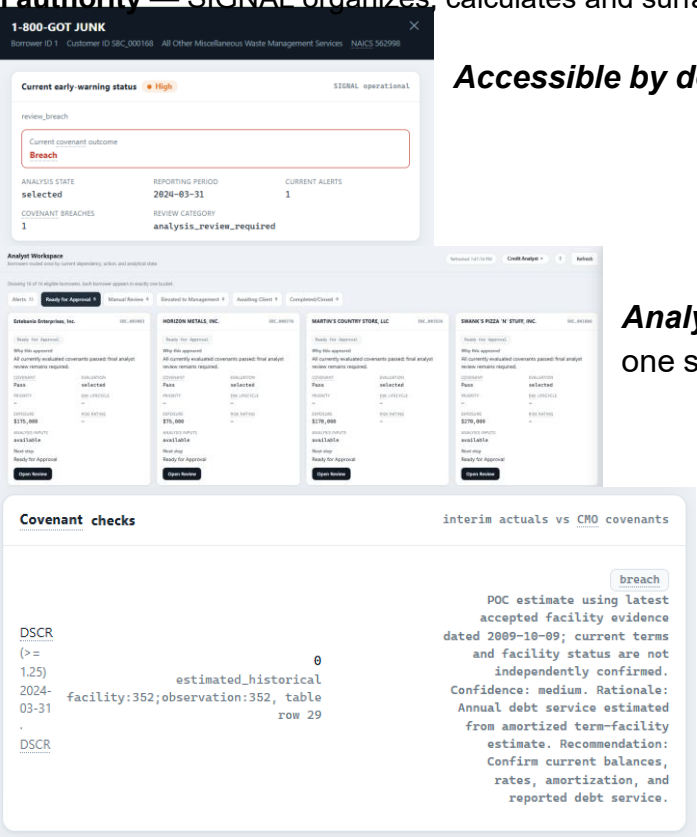
Sees extracted data and analysis, records review decisions.
Read only access to source documents for explainability and transparency.





SIGNAL UX and UI Human Centeredness

Human authority — SIGNAL organizes, calculates and surfaces; qualified people interpret and decide.



Accessible by design: Status conveyed by text + colour; keyboard/focus/contrast considerations.

Analyst agency: Workflow organization supports different working styles rather than prescribing one sequence.

Evidence before automation: Analysts can inspect source evidence behind extracted values and conclusions.

This is *Accessibility-aware design*; not a *claim of formal WCAG certification*.





Security, Privacy & Governance



Data privacy

Synthetic/sample data only. No confidential ATB or customer data.



Access concept

Role-based dashboard navigation across three roles.



Auditability

job_id, source, validation status, and rule/model version stored for every record.



Explainability

Every result shows its formula, threshold, actual value, and source evidence.



Human oversight

Validation status and decision-support labels on every AI/EWI output page.



Version control

Code, rules, data dictionary, and architecture versioned in Git/GitHub.





Why It Matters

Analysts

Less time on manual data extraction and threshold-checking, more time on judgment and borrower engagement.

The Institution

Earlier detection preserves options; a repeatable process is easier to defend to regulators; monitoring scales without headcount growth.

Borrowers

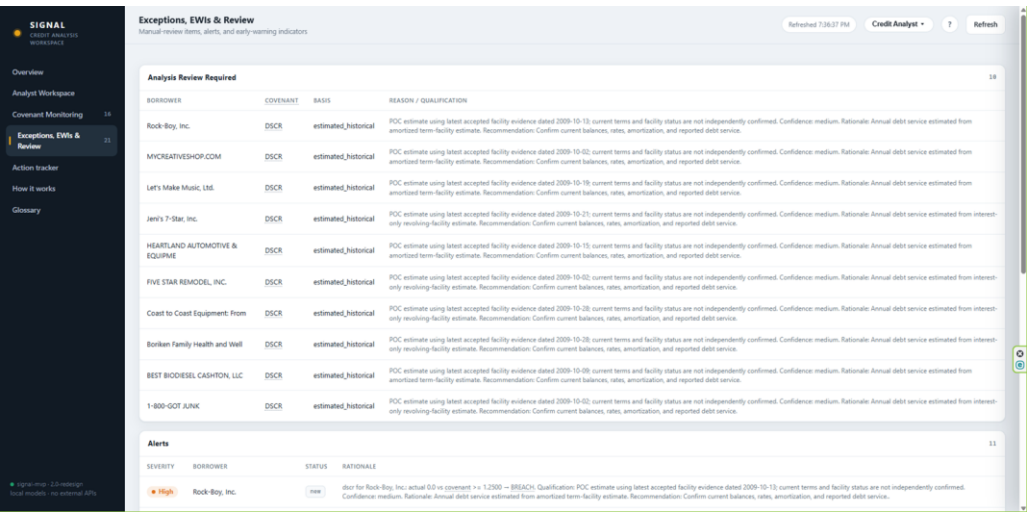
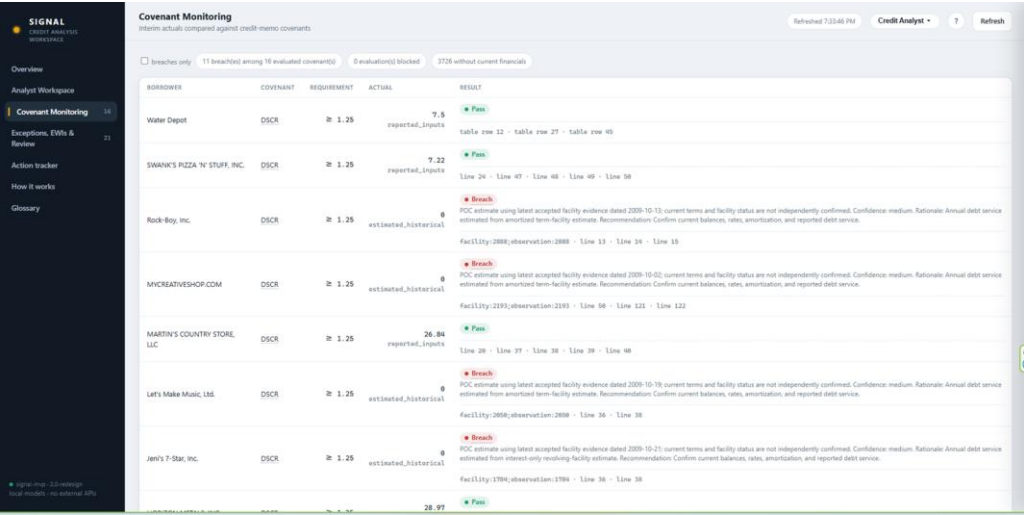
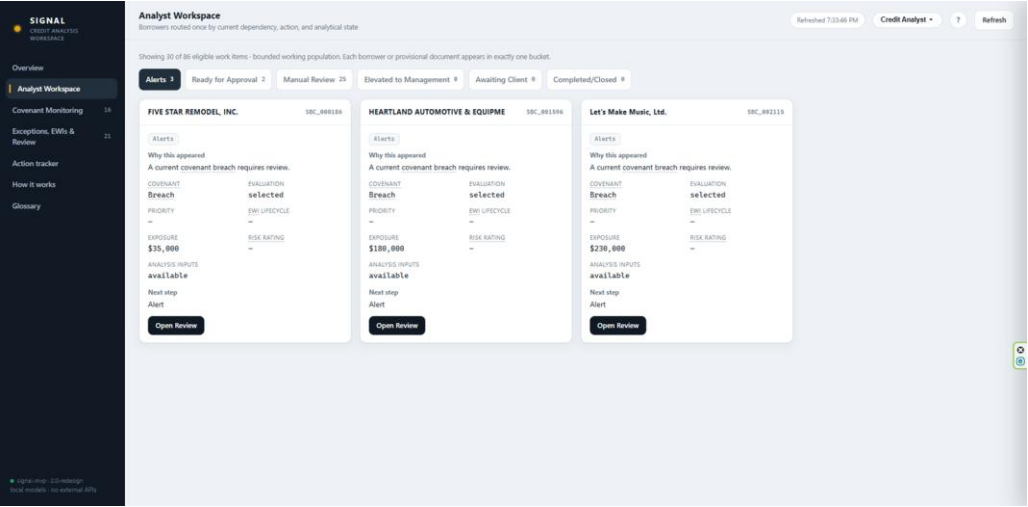
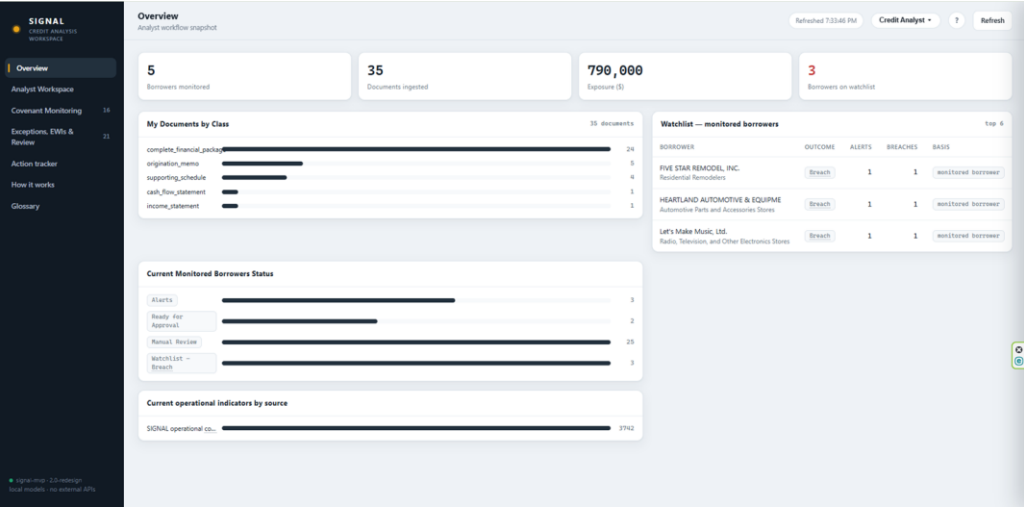
Earlier, more constructive engagement before a financial situation becomes critical.

The Industry

A working example of AI adopted responsibly in credit risk — human oversight, full explainability, no automated adjudication.



05 · STATUS



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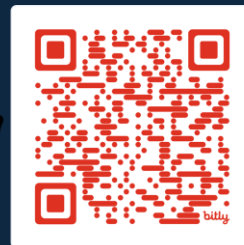
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Thank You

Questions & Discussion

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Academic Advisor: Jaspreet Gill · SAIT PROJ 407 · August 2026



Scan it!



Was the accuracy claim tested fairly?

Model & evidence validation — how SIGNAL checked its own GLiNER extraction model before trusting the numbers.

01

Train on SIGNAL's data

The model was fine-tuned on labeled financial documents so it learns to find the right borrower/company name.

02

Lock away 97 test documents

These were never used in training or tuning — kept aside specifically so the score can't be inflated by memorization.

03

Score once, at the end

Both versions were run on that same untouched set, one time, for a fair comparison.

BEFORE vs. AFTER TRAINING • held-out test set

OUT-OF-THE-BOX

88%

correct company found

~9 wrong guesses per 100 docs

FINE-TUNED ON SIGNAL DATA

98%

correct company found

0 wrong guesses on this test set

A confidence-threshold tweak alone (no retraining) did not improve results — confirming the gain came from training, not a shortcut.

WHAT THIS DOESN'T PROVE YET

All 97 test documents contained a real company name. None were “nothing to find here” cases — so false-alarm behavior on empty documents is still untested, and is disclosed as the next step rather than hidden.



Model complexity follows the business task: interpret where needed, stay deterministic where rules must be reproducible.

CLASSIFY → EXTRACT → APPLY RULES → REVIEW WHEN NEEDED



WHY NOT AN LLM?

Bounded tasks
Classification + span extraction already address the demonstrated need.

Deterministic stages
Known calculations and policy logic stay reproducible and defensible.

Add complexity later
LLMs may fit future open-ended reasoning, synthesis or summarization.

SPECIALIZED NER WAS SUFFICIENT

Zero-shot F1 **0.890** ➔ Fine-tuned F1 **0.990**

97 positive held-out documents • 95/97 borrower entities correctly identified

RIGHT TOOL FOR THE TASK

Interpretation ➔ learned NLP Known patterns / policy ➔ deterministic logic Ambiguity ➔ human review

The learned components provide evidence; downstream rules stay reproducible. Unresolved evidence can remain unresolved rather than forcing an inferred answer.